

Study on the Influential Mechanism of Fluorination on the Flashover Voltage of Polypropylene Film Under Nanosecond Pulse Voltage

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Abstract—A polypropylene (PP) film is usually employed as the main insulation of the peaking capacitor, which is important in an electromagnetic pulse simulator to generate high-amplitude nanosecond pulse. The flashover voltage of PP films under nanosecond pulse limits the highest working voltage of a peaking capacitor. Surface fluorination was employed to increase the flashover voltage of PP. The flashover voltages of PP samples with different fluorination times (0, 4, and 6 h) were tested in an oven with nanosecond pulse voltage. The results showed that the flashover voltage of the PP film with 6-h surface fluorination was increased about 15% than the untreated film in the air. To clarify the influential mechanism, the physicochemical properties of PP film samples were tested and characterized. The results of thermally stimulated depolarization current (TSDC) showed that the trap density of the PP film increased obviously after fluorination treatment. With the increase in treatment time, the trap depth and density were increased obviously. The surface conductivity of PP films slightly decreased with the increase in fluorination time. The mechanism of surface fluorination on the influence of the flashover performance of the PP film under nanosecond pulse is discussed. It is concluded that surface fluorination significantly increases the carbon–fluorine (C–F) bond of the PP film, resulting in the increase in the depth and density of the surface trap, and then, a large amount of homocharges trapped in the surface trap reduces the electric field near the electrode.

Index Terms—Flashover voltage, nanosecond pulse voltage, polypropylene (PP), surface fluorination, thermally stimulated depolarization current (TSDC).

Manuscript received 11 January 2022; revised 20 April 2022; accepted 30 May 2022. Date of publication 3 June 2022; date of current version 4 August 2022. This work was supported by the National Nature Science Foundation of China through the Project Study on the Charge Transport Characters in Composite Insulations Under Liquid Nitrogen Temperature Range and High Temperature Gradient under Grant 52007142. (Corresponding author: Zepeng Lv.)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TDEI.2022.3179993>.

Digital Object Identifier 10.1109/TDEI.2022.3179993

I. INTRODUCTION

THE electromagnetic pulse simulator is an important equipment to produce various specific electromagnetic environments for the research of electromagnetic pulse. The electromagnetic pulse simulator is usually composed of a pulse source and an antenna [1]. The pulse source is composed of a Marx generator, a storage unit, and a peaking unit. The peaking unit consists of a peaking capacitor and an output switch [2]. A peak capacitance is one of the key components of the peak cell and the whole electromagnetic pulse simulator. Polypropylene (PP) is a kind of material with excellent insulation performance and usually used as the main insulation of the peaking capacitor [3], [4]. The flashover voltage of insulation interface is usually lower than the bulk breakdown voltage of the insulation. The surface flashover voltage of PP films under nanosecond pulse is an important factor that restricts the working voltage of peak capacitors [5]. It has been reported that the surface fluorination can effectively increase the flashover voltage of insulation materials under different gas and voltage conditions [6]. An and others studied the effect of fluorination treatment on the electrical properties of epoxy resin insulators. They found that the fluorination treatment effectively increased the dc flashover voltage of epoxy insulators in an SF₆ atmosphere. The mean flashover voltage for 2-min duration of the voltage step increased by 13.6% after the fluorination [7]. Besides, he studied the surface physical and chemical properties and basic properties of the fluorinated PP film. Among them, the dc breakdown strength measurement results showed that the fluorination treatment significantly increased the dc breakdown field strength of the PP film (increased by 23.4%) [8]. Du *et al.* studied the effect of direct fluorination on the surface charge and flashover characteristics of PP films under microsecond pulse voltage. The flashover voltage increases with the increase in the fluorination time of PP films (the flashover voltage under the negative power of 30 min of fluorination is increased by about 3% compared with the non-fluorinated treatment). The flashover voltage under negative power does not decrease with the increase in pulse frequency [9]. So far, the characters of

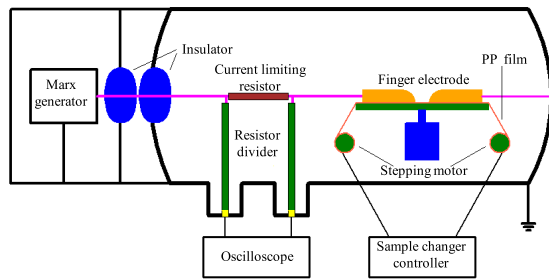


Fig. 1. Main components of the experimental platform. Accessories not shown in the figure include the vacuum pump, the SF_6 recovery unit, and the gas pressure controller.

fluorinated PP sample under nanosecond pulse voltage have not been reported.

In this research, the flashover voltages of PP samples with different fluorination times (0, 4, and 6 h) were tested in an oven with nanosecond pulse voltage. To clarify the influential mechanism, the electrical and chemical properties of PP film samples were tested and characterized. It should be noted that the surface trap density is considered to be an important factor, and the existing methods are not efficient to obtain the surface trap distribution on very thin sample. So, a new method of estimating the surface trap distribution is developed based on the thermally stimulated depolarization current (TSDC). The influential mechanism of fluorination on the flashover voltage of PP film was then discussed.

II. EXPERIMENTAL SETUP

A. Sample Preparation

PP films with a thickness of $18\ \mu\text{m}$ were evenly divided into three parts, two of which were placed into the fluorination apparatus. The mixture of nitrogen and fluorine is used for a fluorination process. The ratio of nitrogen and fluorine is 5:1. The ambient temperature is $35\ ^\circ\text{C}$, and the pressure of fluorination is 20 kPa. The fluorination time is set to be 4 and 6 h. As the sample is so thin, a higher temperature may crimp the film, so that lower temperature and longer fluorination time are set in this article.

B. Surface Flashover Test

The flashover voltages of PP samples with different fluorination times (0, 4, and 6 h) were tested with nanosecond pulse voltage in the air and SF_6 . The experimental platform is shown in Fig. 1. It is mainly composed of a Marx generator, a high-pressure vessel (withstand up to 2 MPa), a controller for the Marx generator, a sample changer, and measurement equipment.

The two polytetrafluoroethylene (PTFE) insulators can withstand a pulse voltage up to 500 kV in the air and SF_6 , and the high voltage pulse is transmitted from the Marx generator to the experimental vessel. The measuring resistor divider, the experimental electrodes, and the sample changer are all placed in the experimental vessel. The distance between the finger electrodes is 5 mm, and the average du/dt of applied nanosecond pulse is 10.15 kV/ns. The ambient temperature of

flashover vessel is $20\ ^\circ\text{C}$, and the humidity of the air is about 10%. The pressure in the vessel is equal to 101 kPa. Other details can be found in [10].

C. Characterization of the Morphology

An atomic force microscope and an optical microscope were employed to characterize the morphology of the fluorinated PP sample. During the test process, the samples to be tested were placed on the glass slide, and then, the slide and the sample were directly placed on the platform of the optical microscope for observation. The appropriate magnification was selected to record the representative morphological features.

The surface roughness of PP samples with different fluorination times was measured by an atomic force microscope. The three kinds of thin-film samples were cut to about $8 \times 8\ \text{mm}$ in size, and the cut samples were glued on a metal sheet with a radius of 5 mm. The metal sheet with samples directly fixed on the platform of an atomic force microscope. A number of selected $20 \times 20\text{-}\mu\text{m}$ areas on the surface of the PP samples with different fluorination time were test.

D. Infrared Spectrum

The chemical functional groups on the surface of the material can be inferred from the results of infrared spectrum. The sample was cleaned to remove the foreign matters on the surface by spraying anhydrous ethanol and wiping with a non-woven cloth. Since the thicknesses of PP samples were small, the film samples were folded three times to improve the accuracy of test results. The infrared spectrum of PP film samples with different fluorination times (0, 4, and 6 h) was tested in a reflection mode. Because the infrared spectrum is measured in the air, it is necessary to use the reference background to eliminate the influence of air on the measurement results.

E. Broadband Dielectric Spectrum

The model of the broadband dielectric spectrometer is Concept 80. Anhydrous ethanol was sprayed on the surface of the samples, and samples were wiped with a non-woven cloth. The PP samples were cut into 15-mm-diameter disks, and gold electrodes were coated on both sides of the sample. The diameters of the coated electrodes are separately 11 and 13 mm for the two sides. The test frequency range is $10^6\text{--}2 \times 10^9\ \text{Hz}$, and the logarithm of frequency sequence is an arithmetic progression.

F. Surface Conductivity

The PP films with different fluorination times (0, 4, and 6 h) were cut into circular samples with a diameter of 50 mm. Anhydrous ethanol was sprayed on the samples surface, and samples were wiped with a clean non-woven cloth to remove the foreign matters on the surface.

A KEITHLEY 6517B programmable electrometer and a model 8009 test electrode system of conductivity were used to test the surface conductivity of samples. To obtain the surface conductivity at different voltages, the applying voltages are,

respectively, set as 50, 200, 500, and 1000 V. The applied voltage polarity was alternatively turned in every 60 s. After several cycles of voltage polarity reversal, the measured current at the end of each voltage application tended to be stable. Then, current readings at 60 s of each voltage application were collected, and the first three readings were abandoned. The average value of the other eight readings was taken as the final result.

III. NEW METHOD OF ESTIMATING THE SURFACE TRAP DENSITY AND DEPTH WITH TSDC

It has been reported that the fluorination can increase the surface trap density and trap depth. The increased deep traps capture more space charge and decrease the electric field near the electrode. The trap density and depth are usually measured and calculated with a surface potential attenuation method. Surface potential attenuation is usually used for thin film or sheet samples with a thickness of μm to mm [11], [12]. One side of the sample is connected to the grounded metal electrode, and the other side is used as the surface to be tested. The surface to be measured is charged by corona charging, electron beam irradiation, or contacting the injection port [13], and then, the decay law of its surface potential is measured. Sjostedt *et al.* [11] analyzed the surface potential of the corona-charged surface of the samples of ethylene-propylene-diene monomer (EPDM) and silicone rubber and measured the distribution information of the charge traps in the material. Min *et al.* [14] proposed the isothermal surface potential decay (ISPD) model and calculated the trap density of polyimide to be $1.3 \times 10^{21} \text{m}^{-3}$. Zhao *et al.* [15] measured the trap parameters of epoxy and aluminum hydroxide composites by surface potential, and their flashover performance under microsecond pulses. This method evaluates the charge transport rate according to the attenuation rate of the surface potential and then obtains the trap parameters. However, if the sample is too thin, more charge passes through the bulk of the insulation rather than through the surface. It will cause great error and underestimate the values of the trap density and depth. So, a method of estimating the trap density and depth based on TSDC is developed and described in the following. The traditional TSDC system can only measure the bulk current of a film, and it is costly to build a new system for surface current measurement. To measure the surface current properly with the TSDC device, surface electrodes should be first made on the surface of the sample, as shown in Fig. 2; then, the sample should be bent to fit the electrode system of TSDC, as shown in Fig. 3. A base insulation plate is used to further reduce the bulk current. So, this method can precisely obtain the surface current during depolarization and calculate the surface trap density with the theory of TSDC.

The TSDC was usually employed to calculate the energy distribution of traps inside insulation materials. So, the electrodes of commercial test device are put at the two faces of the flat sample to get the bulk current. As the surface current is expected to be detected and the electrodes position cannot be changed, the sample shown in Fig. 2 is formed. As shown in Fig. 2, gold electrodes were coated on one side of the

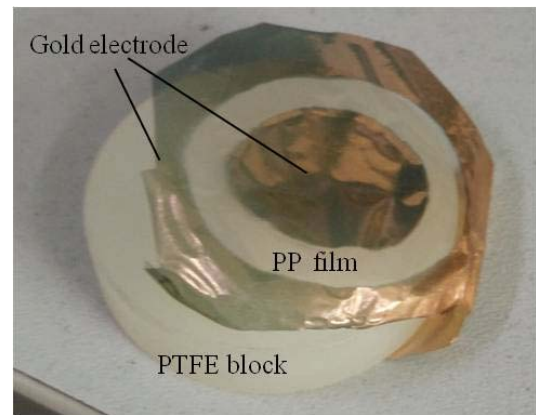


Fig. 2. Photograph of the sample for TSDC.

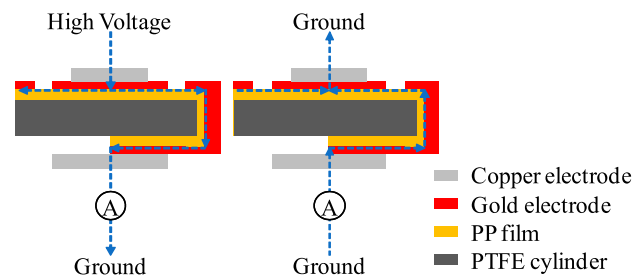


Fig. 3. Diagram of current flow direction.

sample. The coated sample is put upon and bent around a 5-mm-thick PTFE block with its “tail” fixed under the PTFE block, as shown in Fig. 3. The diameter of center electrode is 30 mm. The inside diameter of the outer electrode is 40 mm, and the gap distance between the two electrodes is 5 mm. The high voltage is applied upon the center electrode, and the outer electrode is connected with the electrometer. The thick PTFE block can greatly reduce the current passing through the bulk of insulation. So, the depolarization current of this sample can be considered mainly formed by surface current passing through the gap between the center and outer electrodes.

Fig. 3 shows the flow direction of main current during the polarization (left) and depolarization (right). The orange parts are the sample film. The red parts are coated golden electrodes. The dark gray parts are the PTFE cylinder. The blue arrow shows the current flow direction. When the voltage is applied to the upper electrode, the current can either flow through the surface of the sample or pass through the bulk of cylinder to reach the counter electrode. As the cylinder is very thick, the current passing through the bulk of the cylinder can be neglected. So, the measured data are the surface current passing through the gap between center and outer electrode at the sample surface.

Fig. 4 shows the test procedure of TSDC. The test procedure refers to the plan in [16]. The sample was first polarized by 200 V at 80 °C for 20 min. Then, the temperature was decreased with the cooling rate of 10 °C/min. When the temperature reached −80 °C, the voltage was turned off. After 10 min of surface charge release at −80 °C, the temperature was increased with the heating rate of 2 °C/min, and heating end temperature is 125 °C, as the blue line shown in Fig. 4.

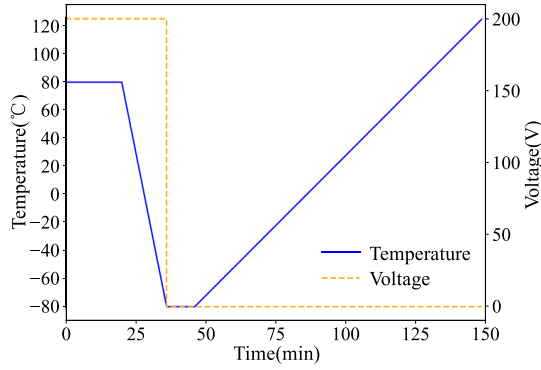


Fig. 4. Test scheme of TSDC.

To simplify the calculation, several hypotheses are made. First, in the polarization stage, it is considered that the traps are homogeneously distributed on the film surface, and the traps capture the charge carriers, which are with lower energy than the trap depth. Second, in the depolarization stage, the energy of the trapped charge increases with the increase in temperature. Once the trapped charge energy is higher than the trap depth, the charge detraps immediately, moves to the electrode, and contributes to the depolarization current.

The energy level of traps is either considered to be discrete or continuous. If the activation energy of carrier traps is considered discrete, the initial rise method, peak temperature method, and curve fitting methods were usually employed to obtain the energy distribution of traps. However, it is indicated that the results calculated by these methods were remarkably less than the true value. The errors illustrated that the energy level of the carrier traps in polymers should be continuous rather than discrete as stated earlier [17]. So, in this article, the energy level of traps was considered to be continuous.

$G(E, T)$ characterizes the charge release proportion in the carrier traps with the activation energy E when the temperature is T . ν is commonly called the attempt-to-escape frequency or frequency factor. It is related to the material to be measured, and it can be taken as 10^{12} Hz for the PP film [18]. $B = 2$ K/min is heating rate

$$G(E, T) = \nu e^{-\frac{E}{kT}} e^{-\frac{1}{\beta} \int_{T_0}^T \nu e^{-\frac{E}{kT}} dT}. \quad (1)$$

$N_t(E)dE$ expresses the number of carrier traps whose the activation energy is from E to $E + dE$ in per unit area. A , area between two surface electrodes, is area without spraying gold. The diameter of center electrode is r_3 , and inside diameter of the outer electrode is r_4 . $AN_t(E)dE$ shows the number of carrier traps corresponding to activation energy interval $[E, E + dE]$. When the sample temperature is T , $AN_t(E)G(E, T)dE$ stands for the number of released charge from the carrier traps with the range of activation energy from E to $E + dE$ on the sample surface in unit time. E_c is the maximum activation energy of the trap, and E_v is the minimum activation energy of the trap. $\int_{E_v}^{E_c} AN_t(E)G(E, T)dE$ expresses the number of released charge from all carrier traps in the area between two surface electrodes of a sample in unit time. The charge quantity of per released charge is

$1.6 \times 10^{-19}C$, which is the electronic charge quantity. The product of the number of released carriers and the charge quantity in unit time is the TSDC. $I(T)$ is the TSDC. Based on the abovementioned analysis, $I(T)$ can be obtained by (3)

$$A = \pi(r_4^2 - r_3^2) \quad (2)$$

$$I(T) = \int_{E_v}^{E_c} eAN_t(E)G(E, T)dE. \quad (3)$$

When the trap density $N_t(E)$ is solved iteratively, the trap density $N_t^{(k)}(E)$ is obtained after the k times iteration. When the sample temperature $T = T_i$, the energy level is $G(E, T_i)$. According to $N_t^{(k)}(E)$ and (3), the TSDC of the sample is recorded as $I^{(k)}(T_i)$. The expression of $I^{(k)}(T_i)$ is shown as follows:

$$\begin{aligned} I^{(k)}(T_i) &= \int_{E_v}^{E_c} eAN_t^{(k)}(E)G(E, T_i)dE \\ &= eA \int_{E_v}^{E_c} N_t^{(k)}(E)G(E, T_i)dE. \end{aligned} \quad (4)$$

If the trap density $N_t^{(k)}(E)$ converges to the true value $N_t(E)$ after k iterations, the result of the trap density $N_t^{(k+1)}(E)$ after $k+1$ iterations also converges to the true value $N_t(E)$. Furthermore, it can be deduced that the thermal depolarization current $I^{(k)}(T_i)$ calculated according to the trap density $N_t^{(k)}(E)$ and the thermal depolarization current $I^{(k+1)}(T_i)$ calculated according to the trap density $N_t^{(k+1)}(E)$ both converge to the measured value $I(T_i)$. Therefore, (5) can be obtained, and the iterative solution formula of trap density distribution $N_t(E)$, as shown in (6), can be obtained by the deformation of (5)

$$\frac{N_t^{(k+1)}(E)}{N_t^{(k)}(E)} = \frac{I(T_i)}{I^{(k)}(T_i)} = 1 \quad (5)$$

$$N_t^{(k+1)}(E) = \frac{N_t^{(k)}(E)I(T_i)}{I^{(k)}(T_i)} = \frac{N_t^{(k)}(E)I(T_i)}{eA \int_{E_v}^{E_c} N_t^{(k)}(E)G(E, T_i)dE}. \quad (6)$$

When (6) is used to solve the trap density $N_t^{(k+1)}(E)$ iteratively, it is necessary to know the trap density after the k iterations and record it as $N_t^{(k)}(E)$. TSDC $I^{(k)}(T_i)$ also need to know. When calculating the TSDC $I^{(k)}(T_i)$ according to (4), we need to know the proportion of charge detrapping in the trap $G(E, T)$, the maximum activation energy level E_c , and the minimum activation energy level E_v of the sample surface trap. According to the TSDC, the range of trap energy levels that can be solved is directly related to the starting and terminal temperatures of the heating and depolarization process in the test scheme and can be got according to the following methods.

$G(E, T)$ is the percentage of charge detrapping from the trap with E energy level at temperature T . $G(E, T)$ is a function of one variable of the activation energy E when the temperature $T = T_i$ remains constant. The ratio of charge trapping in different energy level traps is all larger than zero, and the unique maximum $G(E_0, T_i)$ is obtained when $E = E_0$. $G(E_0, T_i)$ is also the maximum of $G(E, T_i)$. When the temperature $T = T_i$, there exists an interval $[E_1, E_2]$ that satisfies (7). The values of E_1 and E_2 are dependent on the temperature T_i and positive integer n . When the sample temperature is

$T = T_i$, because most of the charges in the traps with the less activation energy have detrapped at a lower temperature, the charges in these traps will detrapp very little. The trap with higher activation energy has not been activated at the sample temperature T_i , and the charge that can escape from the trap is very little. Therefore, when n is a large positive integer, the value of $G(E, T_i)$ outside the interval $[E_1, E_2]$ is obviously smaller than the value of the interval $[E_1, E_2]$, and the part outside the interval $[E_1, E_2]$ can be ignored. Because the charge in the trap with lower activation energy at this initial temperature can be considered to have been almost completely released when the surface charge is released by short circuit, the starting point of the interval E_1 corresponding to the initial temperature T_0 is taken as the minimum activation level E_v . Because the traps with higher activation energy are not activated when the sample reaches the terminal temperature, the charge in these traps can hardly escape from the trap and, thus, form the thermal stimulation depolarization current. So, the terminal point of the interval E_2 corresponding to the end temperature is used as the maximum activation level E_c .

$$G(E_1, T_1) = G(E_2, T_1) = \frac{1}{n} G(E_0, T_1) E_0 \in [E_1, E_2]. \quad (7)$$

The solution of the trap level distribution on the sample surface can be completed in the following steps. In the first step, according to the experimental scheme and material characteristics, the charge detrapping ratios of different activation energy traps at different temperatures are obtained according to (1). The second step is to select a larger positive integer n according to the different activation energy trap charge detrapping ratios and test schemes and to calculate the starting point and ending point of activation energy range corresponding to each test temperature point T_i according to (7). The third step is to set the initial value of surface trap energy level density distribution according to the corresponding activation energy range of each temperature and then set the initial value of i as 1. The fourth step is to calculate the TSDC $I(T)$, when $T = T_i$ by substituting the current trap energy level density distribution and trap charge detrapping ratio into (4). The fifth step is to substitute both measured and calculated data of the TSDC $I(T)$ into (6), calculate the trap level density of each level in the activation energy range $[E_1, E_2]$ corresponding to the temperature T_i , and update the trap level distribution. The sixth step is to judge whether the temperature T_i is the end temperature of the test. If the current temperature point T_i is the end temperature, the seventh step is carried out. Otherwise, increase the value of i by 1 and return to the fourth step. The seventh step is to judge whether the current obtained TSDC makes (5) tenable. If (5) is tenable within the acceptable error range, it means that the density distribution of trap energy level on the sample surface has been obtained, and the whole solving process is ended. Otherwise, change the value of i to 1, return to the fourth step, and continue the iterative solution. The method of the seventh step to judge whether the equation is true is to draw the measured current data and the calculated current data in the same coordinate system for comparison, and compare whether the two curves coincide. Doing a seventh step once is called a total iteration.

TABLE I
WEIBULL DISTRIBUTION PARAMETERS OF THE FLASHOVER VOLTAGES OF PP SAMPLES WITH DIFFERENT FLUORINATION TIMES UNDER NANOSECOND PULSE VOLTAGE IN THE AIR AND SF₆

Parameter	Untreated sample	4 hrs' fluorination	6 hrs' fluorination
$\alpha(\text{air})$	12.53 kV	13.69 kV	14.43 kV
$\beta(\text{air})$	34.14	20.08	83.18
$\alpha(\text{SF}_6)$	21.03 kV	22.25 kV	23.97 kV
$\beta(\text{SF}_6)$	14.40	16.23	16.84

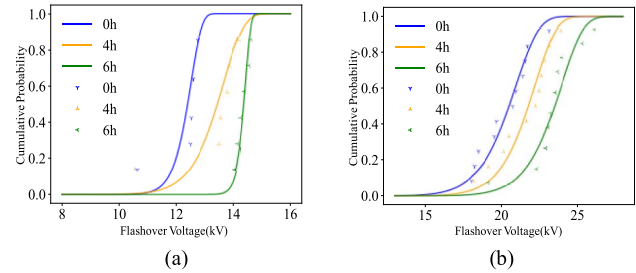


Fig. 5. Probability distribution of flashover voltage of PP samples with different fluorination times under nanosecond pulse. (a) In the air. (b) In the SF₆.

IV. RESULTS AND DISCUSSION

A. Flashover Voltage

The test result of flashover voltage of each sample is usually considered to follow two parameters Weibull distribution. According to the flashover voltages of PP samples with different fluorination times under nanosecond pulse voltage in the air and SF₆, the scale parameter α and shape parameter β of the two parameters Weibull distribution are calculated by the maximum likelihood estimation method. The Weibull distribution parameters of the flashover voltages of different samples are shown in Table I. The cumulative probability of the flashover voltage of the samples in the air and SF₆ was plotted in Fig. 5. The characteristic flashover voltages in the air of the samples with 0-, 4-, and 6-h fluorination are, respectively, 12.33, 13.33, and 14.33 kV. The flashover voltage of the PP film with 6-h fluorination increased about 15% than the untreated film in the air. The characteristic flashover voltage in the SF₆ of the samples with 0-, 4-, and 6-h fluorination is, respectively, 20.28, 21.54, and 23.23 kV. The flashover voltage of the PP film with 6-h fluorination increased about 14% than the untreated film in the SF₆. For both of the tests in the air and SF₆, the fluorinated samples show a higher flashover voltage. The longer the fluorination time, the samples have the higher flashover voltage.

B. Characterization of the Morphology

The surface morphologies of three kinds of samples were first observed under an optical microscope. The photographs of the surface topography of the three samples with 100 times magnification are shown in Fig. 6. It is usually considered that the chemical bond breaking and exothermic phenomenon during the fluorination process may change the surface roughness. However, it is hard to see obvious difference among the

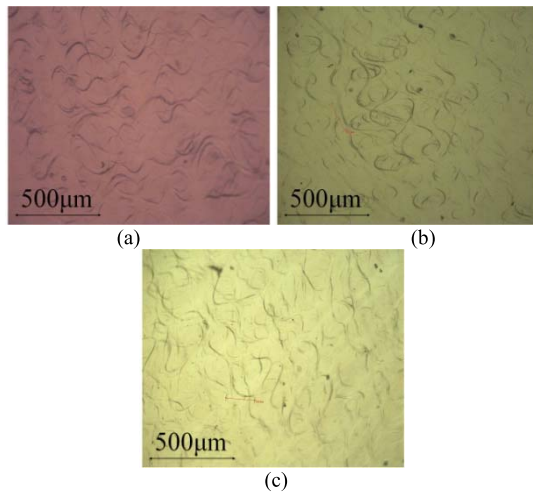


Fig. 6. Surface morphology of PP samples with different fluorination times. (a) Untreated sample. (b) 4-h fluorination. (c) 6-h fluorination.

samples shown in Fig. 6. It maybe because the original PP film is roughened before, and its rough is much bigger than the change by the fluorination. The surface topography of those samples should be evaluated by other methods.

To evaluate the roughness of the samples, the atomic force microscopy (AFM) was employed. The surface height data of PP samples were measured. The measurement of each face of the sample repeated at least six times to test the uniformity of the roughness. It is found that one side of PP film is smooth, and the roughness is between 20 and 50 nm; the other side is rougher, and the roughness is between 50 and 220 nm. However, as shown in Fig. 7, it seems that the change of roughness caused by fluorination is lower than 40 nm and cannot significantly change the overall roughness of the sample, especially the origin PP sample has already been roughened before.

C. Infrared Spectrum

The infrared spectra of three kinds of samples were tested by the reflection mode. The test results of infrared spectrum were plotted in the form of absorbance in Fig. 8 (as shown in the red dotted box in Fig. 8). The infrared spectrum of fluorinated PP samples showed obvious change in the wavenumber range of 1100–1350 cm^{-1} and the range of 2700–3000 cm^{-1} . The characteristic absorption peak in the wavenumber range of 1100–1350 cm^{-1} is corresponding to the existence of carbon–fluorine (C–F) bonds. There is an obvious characteristic absorption peak in the wavenumber range of 2700–3000 cm^{-1} , which is corresponding to the saturated hydrocarbon (C–H) bonds [19]. It can be seen that the absorption peak in the range of 2700–3000 cm^{-1} decreased with the fluorination time, showing that the fluorination process decreased the number of C–H bond. The sample without fluorination did not have obvious absorption peak in the range of 1100–1350 cm^{-1} ; with the increase in the fluorination time, the peak in the range of 1100–1350 cm^{-1} increased sharply, which showed that the fluorination process increased the number of C–F bond. Then, it can be concluded that

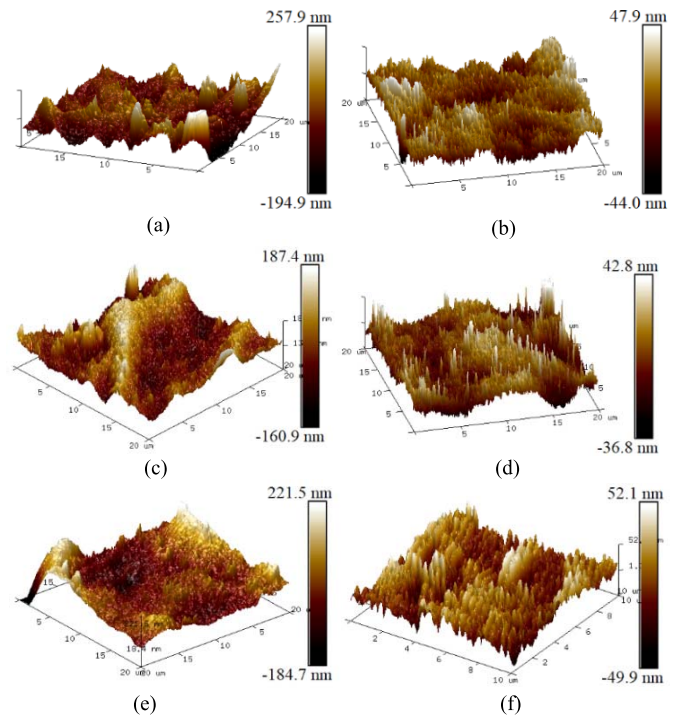


Fig. 7. AFM images of PP samples with different fluorination times. (a) Untreated sample-rough surface (RF). (b) Untreated sample-smooth surface (SF). (c) 4-h fluorination-RF. (d) 4-h fluorination-SF. (e) 6-h fluorination-RF. (f) 6-h fluorination-SF.

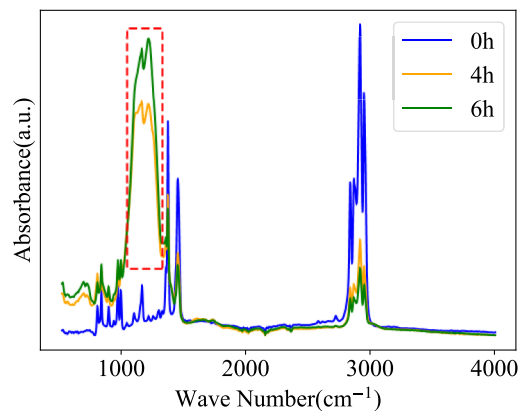


Fig. 8. Infrared spectrum of PP samples with different fluorination times.

the fluorination process replaces hydrogen atoms with fluorine atoms and forms C–F bond; the longer the fluorination time, the more C–H bonds are turned into C–F bonds.

D. Broadband Dielectric Spectrum

Fig. 9 shows the test result of the broadband dielectric spectrum. The relative permittivity of fluorinated PP increases slightly. The relative permittivity of PP remained stable in the range from 10^6 to 10^9 Hz. When the PP film is used as the main insulating material of peaking capacitor, the relative permittivity of the PP film is closely related to the energy storage performance of the peaking capacitor. When other parameters of the peaking capacitor are the same, the larger the relative permittivity of the PP film is, the larger the

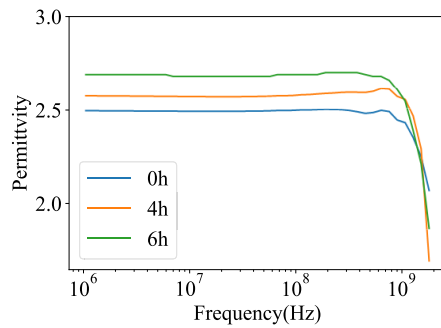


Fig. 9. Broadband dielectric spectrum of PP films with different fluorination times.

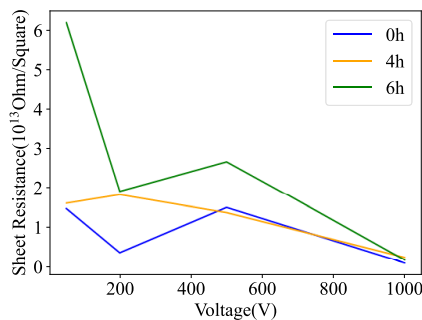


Fig. 10. Surface conductivity of PP films with different fluorination times.

capacitance of the capacitor is and the better the energy storage performance is. According to the test results of the dielectric spectrum of the three PP samples, fluorination treatment is beneficial to improve the energy storage performance of the peaking capacitor. Another point should be pointed that the relative permittivity of all the samples drops sharply when the frequency is higher than 10^9 Hz. The average du/dt of applied nanosecond pulse used in this article is 10.15 kV/ns. The rising edge of the nanosecond pulse is 1–2 ns. So, the effect of the frequency higher than 10^9 Hz can be neglected.

E. Sheet Resistance

The sheet resistances of PP samples with different fluorination time are shown in Fig. 10. It can be seen that with the increase in the fluorination time, the sheet resistance of PP samples increases. It is usually considered that the increase in the sheet resistance will make it difficult for the surface charge to dissipate in time, which is not effective to improve the flashover voltage [20], [21]. Therefore, it seems that the change of sheet resistance of PP films due to fluorination may not be the main reason for the improvement of flashover voltage under nanosecond pulse.

F. Thermally Stimulated Depolarization Current

The TSDC of the samples with different fluorination time is shown in Fig. 11. It can be seen that the depolarization currents of fluorinated samples are much bigger than that of the untreated sample. With the increase in fluorination time, the depolarization current increased. The 6-h fluorinated sample had the highest peak current, which is ten times higher

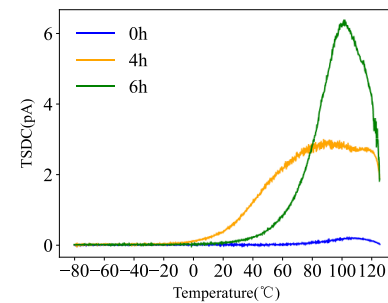


Fig. 11. TSDC of PP films with different fluorination times.

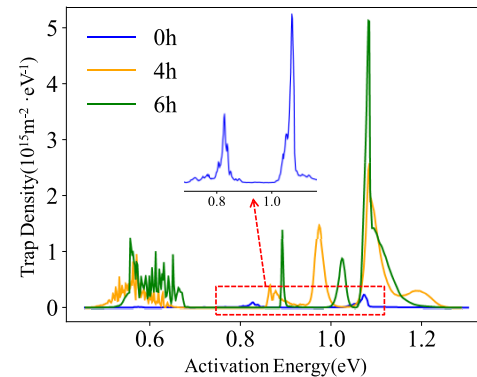


Fig. 12. Density distribution of trap energy level of PP films with different fluorination times.

than that of the untreated sample. It should be noted that the depolarization current of the 6-h fluorinated sample is lower than that of the 4-h fluorinated sample when the temperature is lower than 80 °C. It indicated that the trap of the 6-h fluorinated sample is deeper, and that the trapped charge cannot detrapp at low temperature.

Based on the measured data of the thermal stimulation depolarized surface current, the density distribution of the sample surface trap is calculated by the method described earlier. After several iterations, the surface trap energy density distribution of the PP film with different fluorination times was calculated according to the method proposed in Section III. Fig. 12 shows the density distribution of trap energy level of PP films with different fluorination times. Since the trap density of the PP sample without fluorination is significantly smaller than those of the fluorinated samples, its local curve is separately shown in Fig. 12.

The increase in surface trap depth and density caused by surface fluorination is closely related to the change of surface chemical composition. As shown by an infrared spectrum, there are a lot of C–F bonds on the surface of fluorinated films. Because of the strong electronegativity of fluorine atom, its attraction to electrons is obviously stronger than that of hydrogen atom and carbon atom, and the polarity of C–F bond is stronger than that of hydrocarbon bond. Since the existence of carbon and fluorine bonds, the fluorinated samples have stronger surface polarity and stronger electron adsorption and capture ability than untreated PP films. Therefore, the surface trap depth of fluorinated samples increases. With the increase in fluorination time, the number of C–F bonds increases,

which leads to the increase in trap density on the surface of fluorinated samples.

In the application of nanosecond pulse voltage, homocharge will first accumulate in the front of electrodes. The ionic species and the injected carriers can hardly pass through the distance between electrodes to form heterocharge in several nanoseconds. As the surface trap depth and density are increased by the fluorination, more injected charge carriers will be trapped near the injection electrode on the surface of fluorinated samples during the period of voltage rising. For one thing, the amount of mobile charge carriers will be decreased. For another, the electric field near the electrode will be decreased. Therefore, the measured sheet resistance of the fluorinated samples increases with the increase in fluorination time and trap density. It is consistent with the experimental results.

The amount of seed electron and electric field strength are two important factors that determine the discharge. When the electric field exceeds a certain threshold under the high voltage, the electron accelerates and collides with the gas under the action of electric field. It is possible to cause secondary electrons. The multiplication of free electrons leads to electron avalanche, which leads to partial discharge and even flashover along the surface. As for the fluorinated samples, the C–F bonds show stronger electronegativity than C–H bonds, and it may prevent the ionization from the surface and reduce the number of secondary electrons [9]. From another perspective, the density and depth of the surface trap increase due to fluorination. More carriers may be trapped and form homocharge in the front of the electrodes, and the amount of mobile carrier decreases in the fluorinated samples. The homocharge reduces the electric field of the electrodes. With both the electric field of electrode and the amount of mobile carrier reduced, the discharge gets hard to be initiated. Therefore, it makes sense that the increase in trap concentration and depth due to the fluorination may be the key factor to improve the flashover performance.

V. CONCLUSION

To improve the flashover voltage of PP under nanosecond pulse voltage, the fluorination was employed. It is found that the flashover voltages of the PP samples under nanosecond pulse voltage in both the air and SF₆ increase with the fluorination time. After 6-h fluorination, the flashover voltage of PP film increased by 15% in the air and 14% in the SF₆.

The physical and chemical properties of surface fluorinated PP were analyzed to verify the influential mechanism of the surface fluorination. Especially, a new method of estimating the surface trap depth and density for thin sample was developed based on TSDC. It is found that with the increase in fluorination time, more C–H bonds were replaced by the C–F bond; the trap depth and density on the surface of the film increased significantly with the fluorination time. The increased trap depth and density due to the surface fluorination will trap more carriers and form homocharge. It then reduces the number of mobile carriers and the electric field

near electrodes. That may be the main reason that the surface fluorination increases the flashover voltage of PP sample under nanosecond pulse voltage.

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